THE MASTERMIND COACHING CLASSES

Revision Notes

Class 7 · Mathematics

Ganita Prakash — Part 2, Chapters 5 to 8 · CBSE / RBSE Syllabus

HOW TO USE THESE NOTES

Each chapter follows the textbook's own order of sections, so you can read straight through. Bold words are keywords you must be able to define. Tables compare and classify; shaded boxes hold definitions, formulas and key facts. The navy Key Takeaways box at the end of every chapter is your last-two-minutes revision.

CH.	CHAPTER TITLE	CORE THEME
5	Parallel and Intersecting Lines	Angle pairs, vertically opposite angles, transversal and parallel lines, corresponding, alternate and co-interior angles
6	Number Play	Parity, number grids, digit tricks, divisibility, Kaprekar and palindrome routines, reasoning
7	A Tale of Three Intersecting Lines	Triangle types, triangle inequality, angle sum and exterior angle, altitudes and medians, constructing triangles
8	Working with Fractions	Multiplying fractions, fraction of a quantity, reciprocals, dividing fractions, word problems

The Mastermind Coaching Classes — Road No. 5, VKIA, Jaipur — 9887258679

CHAPTER 05 · MATHS (PART 2)

Parallel and Intersecting Lines

When lines cross or run side by side they create angles with fixed relationships. This chapter studies pairs of angles (complementary, supplementary, linear pair, vertically opposite) and the angles made when a transversal cuts parallel lines, then uses them to find unknown angles and to prove lines parallel.

Big Questions

- Which angle pairs are always equal or always add to 90° or 180°?
- What angles does a transversal make with two parallel lines?
- How do we use these results to find unknown angles and test for parallel lines?

5.1 Angle Pairs

Pair	Meaning	Measure
Complementary angles	two angles whose sum is 90°	e.g. 35° and 55°
Supplementary angles	two angles whose sum is 180°	e.g. 110° and 70°; two acute angles can never be supplementary
Adjacent angles	share a vertex and an arm, with no overlap	—
Linear pair	adjacent angles whose non-common arms form a straight line	always add to **180°**
Vertically opposite angles	opposite angles formed by two intersecting lines	always **equal**

FORMULA

Angles on a straight line = **180°**. Angles around a point = **360°**. Angles of a right angle = **90°**.

Worked example. Lines intersect making an angle of 70°. Its vertically opposite angle is 70°; the two remaining angles are each a linear pair partner: 180° − 70° = **110°**, 110°. The angles 90°, 120°, 100° and x around a point give x = 360° − 310° = **50°**.

▸ Proof: vertically opposite angles are equal

Let lines AB and CD cross at O. ∠AOC + ∠COB = 180° (linear pair) and ∠COB + ∠BOD = 180° (linear pair). Hence ∠AOC = ∠BOD. In the same way ∠COB = ∠AOD.

5.2 Parallel Lines and a Transversal

DEFINITION

A **transversal** is a line that cuts two (or more) lines at distinct points. Two lines in a plane are **parallel** if they never meet however far extended; the distance between them stays constant.

A transversal cutting two lines makes 8 angles. They form named pairs:

Pair	Position	If lines are parallel
Corresponding angles	same side of the transversal, same relative position (F-shape)	**equal**
Alternate interior angles	opposite sides of the transversal, between the lines (Z-shape)	**equal**
Co-interior (allied) angles	same side of the transversal, between the lines (C or U shape)	**supplementary** (sum 180°)
Vertically opposite angles	across the crossing point	always equal (parallel or not)

5.3 Finding Unknown Angles

Problem	Working	Answer
Corresponding angles (3x + 10)° and (5x − 30)°, lines parallel	3x + 10 = 5x − 30, so 2x = 40	x = 20; angles 70°
Co-interior angles (2x + 15)° and (3x + 5)°, lines parallel	5x + 20 = 180	x = 32; angles 79° and 101°
Alternate angles (2x + 10)° and (x + 40)°, lines parallel	2x + 10 = x + 40	x = 30; angles 70°
∠1 = 52°; find its alternate and co-interior partners	alternate = 52°; co-interior = 180° − 52°	52° and 128°

Step 1 — Mark
Mark the given angles and identify the relation (corresponding, alternate, co-interior, linear pair, vertically opposite).

Step 2 — Equation
Equal pair: set expressions equal. Supplementary pair: set their sum to 180°.

Step 3 — Solve and check
Solve for x, substitute back and check that the angles satisfy the relation.

5.4 Testing for Parallel Lines (Converse)

IMPORTANT

If a transversal makes **equal corresponding angles**, or **equal alternate interior angles**, or **co-interior angles that add to 180°**, then the two lines are **parallel**. The converse is also true.

- A **carpenter's** or **draftsman's** set square slid along a ruler draws parallel lines because it keeps corresponding angles equal.
- Railway tracks, ruled lines and the edges of a road are everyday parallel lines.
- In a clock at 3 o'clock the hands are perpendicular (90°); two intersecting lines making 90° are **perpendicular**.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Define parallel and intersecting lines.	Parallel lines never meet; intersecting lines meet at one point.
What is a linear pair? Its measure?	Adjacent angles on a straight line; they add to 180°.
Prove that vertically opposite angles are equal.	Both make a linear pair with the same angle, so ∠AOC = ∠BOD.
Two intersecting lines make 70°. Find the other angles.	70°, 110°, 110°.
Angles around a point are 90°, 120°, 100° and x.	x = 50°.
What are alternate interior angles?	Angles on opposite sides of the transversal, between the lines; equal if the lines are parallel.
Corresponding angles (3x + 10)° and (5x − 30)° are equal. Find x.	x = 20.
True or false: two acute angles can be supplementary.	False; acute + acute < 180°.
Co-interior angles are 4x and 5x with parallel lines. Find x.	9x = 180, x = 20.
State the converse test for parallel lines.	Equal corresponding or alternate angles, or co-interior angles adding to 180°, make the lines parallel.

★ KEY TAKEAWAYS

- **Linear pair = 180°**; vertically opposite angles are **equal**; angles at a point = **360°**.
- Parallel lines + transversal: **corresponding equal, alternate equal, co-interior supplementary**.
- **Complementary = 90°, supplementary = 180°.**
- Set up an **equation** from the angle relation to find x, then check.
- The **converse** of each result is used to prove that lines are parallel.

CHAPTER 06 · MATHS (PART 2)

Number Play

This chapter treats numbers as a playground. You use parity (odd and even), digits, divisibility tests and patterns such as multiples in a grid, palindromes and Kaprekar's routine to explain why number tricks work, and you learn to prove a statement for all numbers rather than a few examples.

Big Questions

- How does parity (odd or even) help in predicting results?
- What patterns do multiples and digit sums show and why?
- How do we explain a number trick for all numbers, not just one?

6.1 Odd, Even and Parity

DEFINITION

A number is **even** if it is divisible by 2 (it ends in 0, 2, 4, 6, 8) and **odd** otherwise. **Zero is even** because 0 = 2 × 0. Odd or even is the number's **parity**.

Operation	Result	Example
odd + odd	even	7 + 9 = 16
even + even	even	6 + 8 = 14
odd + even	odd	5 + 8 = 13
odd × odd	odd	7 × 9 = 63
any × even	even	13 × 14 × 15 is even (has the even factor 14)
sum of many numbers	odd only if the **count of odd numbers is odd**	2 + 4 + 6 + 8 + 9 is odd (one odd number)

A number written as **2k** is even and **2k + 1** is odd. Using these forms we can prove statements for **every** number: (2a + 1) + (2b + 1) = 2(a + b + 1), which is even.

6.2 Patterns in Grids and Sequences

- Write 1 to 100 in rows of 10. Multiples of **9** lie on a slanting line (9, 18, 27, …, 90) and their digits add to 9.
- Sum of the first n odd numbers = $\mathbf{n^2}$: 1 + 3 + 5 + 7 + 9 + 11 = 36 = 6^2.
- 99 × 99 = 9,801 and 100 × 98 = 9,800: (n − 1)(n + 1) = n^2 − 1, so the square is always **1 more**.
- The sum of the first 8 odd numbers is 8^2 = 64, which is greater than 60.

6.3 Palindromes and Digit Routines

DEFINITION

A **palindrome** reads the same both ways. A 3-digit palindrome has the form aba (121, 454). Palindromes between 50 and 70 are 55 and 66.

Kaprekar's step
Arrange the digits of a 4-digit number in descending and ascending order and subtract. For 3524: 5432 − 2345 = **3087**. Repeating reaches **6174** within 7 steps.

Reverse and add
Add the number to its reverse until a palindrome appears: 68 → 154 → 605 → 1111.

1089 trick
Take any 3-digit number whose first and last digits differ by 2 or more; subtract its reverse and add the result to its reverse: always **1089**.

NOTE

Why does the 1089 trick work? For abc with a > c the difference abc − cba equals 99 × (a − c), which has 9 in the tens place; adding its reverse always gives 1089.

6.4 Divisibility and Reasoning

Test	Rule	Check on 7,284
by 3	digit sum divisible by 3	7 + 2 + 8 + 4 = 21 ✓
by 4	last two digits divisible by 4	84 = 4 × 21 ✓
by 9	digit sum divisible by 9	21 ×
by 11	difference of alternate digit sums is 0 or a multiple of 11	(7 + 8) − (2 + 4) = 9 ×

Checking by reasoning. A number is divisible by 12 exactly when it is divisible by 3 **and** 4 (co-prime factors). A number is divisible by 6 if it is even and its digit sum is a multiple of 3. A single counterexample disproves a claim; a proof must cover all cases.

Quick Revision: Likely Exam Questions

Likely question	Model answer
State the rule for the sum of two odd numbers.	The sum is even, e.g. 7 + 9 = 16.
Why is 0 even?	0 = 2 × 0, so it is divisible by 2.
Is 2,35,716 even or odd?	Even, since it ends in 6.
Which is bigger, 99 × 99 or 100 × 98?	99 × 99 = 9,801 > 9,800.
Apply one Kaprekar step to 3,524.	5432 − 2345 = 3087.
Is 7,284 divisible by 3 and 4?	Yes; digit sum 21 and last two digits 84.
Is 13 × 14 × 15 odd or even? Why?	Even; it has the even factor 14.

Likely question	Model answer
Add 1 + 3 + 5 + 7 + 9 + 11. What pattern?	$36 = 6^2$.
Two palindromes between 50 and 70?	55 and 66.
Using parity, show 2 + 4 + 6 + 8 + 9 is odd.	It contains exactly one odd number.

★ KEY TAKEAWAYS

- **Parity:** odd + odd = even, odd × odd = odd, even × anything = even; **0 is even**.
- Write even as **2k** and odd as **2k + 1** to prove statements for all numbers.
- Sum of the first n odd numbers = $\mathbf{n^2}$; $(n - 1)(n + 1) = n^2 - 1$.
- **Kaprekar** (6174), **reverse-and-add** (palindromes) and **1089** are routines with surprising endpoints.
- Divisibility by a composite number: test its **co-prime factors**; one counterexample disproves a claim.

CHAPTER 07 · MATHS (PART 2)

A Tale of Three Intersecting Lines

Three lines that pairwise intersect enclose a triangle. This chapter classifies triangles, finds when three lengths can form one (triangle inequality), proves the angle sum and exterior angle properties, introduces altitudes and medians and constructs triangles from given measurements.

Big Questions

- When can three lengths or three angles form a triangle?
- What do the angle sum and exterior angle properties say?
- How are altitudes and medians drawn and how do we construct a triangle?

7.1 Types of Triangles

By sides	Description	By angles	Description
Equilateral	all 3 sides equal; each angle 60°	**Acute-angled**	all angles less than 90°
Isosceles	2 sides equal; angles opposite them equal	**Right-angled**	one angle exactly 90°; longest side is the hypotenuse
Scalene	all sides different	**Obtuse-angled**	one angle more than 90°

A triangle has **3 vertices, 3 sides and 3 angles**. Only **one** angle can be right or obtuse, because the angle sum is fixed.

7.2 Triangle Inequality

IMPORTANT

The **sum of any two sides of a triangle is greater than the third side**. Three lengths a, b, c form a triangle only if a + b > c, b + c > a and a + c > b. It is enough to check that the **two shorter sides add to more than the longest**.

Lengths (cm)	Check	Triangle?
4, 5, 6	4 + 5 = 9 > 6	Yes
4, 5, 10	4 + 5 = 9 < 10	No
3, 4, 7	3 + 4 = 7 (not greater)	No: it collapses to a line
5, 12, 13	5 + 12 = 17 > 13	Yes

7.3 Angle Sum and Exterior Angle

FORMULA

Angle sum property: the three interior angles of a triangle add up to **180°**. **Exterior angle property:** an exterior angle equals the **sum of the two interior opposite angles**.

- Proof idea: through one vertex draw a line parallel to the opposite side; alternate angles equal two of the angles, and the three angles together make a straight line (180°).
- A triangle with angles 90°, 60°, 40° is impossible (sum 190°); each angle less than 60° is impossible (sum under 180°).
- In an isosceles triangle with each base angle 4 times the third angle: 9x = 180, x = 20°, so the angles are **20°, 80°, 80°**.
- Angles (x + 10)°, 2x°, (3x − 10)°: 6x = 180, x = 30, angles **40°, 60°, 80°**.
- Exterior angle example: interior opposite angles 50° and 60° give an exterior angle of **110°**.

7.4 Altitudes and Medians

Line	Definition	Special point
Altitude	perpendicular from a vertex to the opposite side (or its extension)	Three altitudes meet at the **orthocentre**
Median	segment from a vertex to the **midpoint** of the opposite side	Three medians meet at the **centroid**, the balance point

NOTE

Every triangle has **three** altitudes and **three** medians. In an obtuse triangle the orthocentre lies **outside** the triangle; in a right triangle it is at the right-angle vertex.

7.5 Constructing Triangles

Given	Key idea
SSS (three sides)	draw the longest side as base; arcs from its ends with the other two lengths meet at the third vertex
SAS (two sides and the angle between them)	draw one side, measure the angle at its end, mark the second side
ASA (two angles and the side between them)	draw the side, make the angles at both ends; the arms meet at the vertex
RHS (right angle, hypotenuse, one side)	draw the leg, erect the right angle, cut with the hypotenuse arc

Draw 4, 5, 6 cm triangle

Draw base 6 cm. With centre at one end, arc of radius 4 cm; from the other end, arc of 5 cm; mark the intersection as the vertex and join.

Right-angle check. Sides 3, 4, 5 satisfy $3^2 + 4^2 = 9 + 16 = 25 = 5^2$, so the triangle is **right-angled** (Pythagoras' relation, studied in detail later).

Quick Revision: Likely Exam Questions

Likely question	Model answer
State the angle sum property.	The three interior angles of a triangle add to 180°.
Can a triangle have sides 4 cm, 5 cm and 10 cm?	No, because 4 + 5 < 10.
Name the types of triangles based on sides.	Equilateral, isosceles, scalene.
How many altitudes does a triangle have? Where do they meet?	Three; at the orthocentre.
Is a triangle with angles 90°, 60°, 40° possible?	No; the sum is 190°.
Define a median.	A segment joining a vertex to the midpoint of the opposite side.
State the exterior angle property.	An exterior angle equals the sum of the two interior opposite angles.
Angles (x + 10)°, 2x°, (3x − 10)°. Find x.	x = 30.
Is a triangle with sides 3, 4, 5 right-angled?	Yes; 9 + 16 = 25.
Is every angle less than 60° possible in a triangle?	No; the sum would be below 180°.

★ KEY TAKEAWAYS

- Triangle types: **equilateral, isosceles, scalene** (sides) and **acute, right, obtuse** (angles).
- **Triangle inequality:** two shorter sides must add to more than the longest.
- **Angle sum = 180°**; exterior angle = sum of the two **opposite interior** angles.
- **Altitude** is perpendicular to the side; **median** goes to the midpoint; each triangle has three of each.
- Construct a triangle from **SSS, SAS, ASA or RHS** data.

CHAPTER 08 · MATHS (PART 2)

Working with Fractions

Last year you added and subtracted fractions. Now you multiply and divide them. This chapter explains multiplication as 'a fraction of a fraction', introduces reciprocals, shows why dividing by a fraction means multiplying by its reciprocal and applies both operations to measurement and sharing problems.

Big Questions

- What does it mean to multiply by a fraction and how is it done?
- What is a reciprocal and how do we divide by a fraction?
- How do these operations solve everyday problems?

8.1 Multiplying Fractions

FORMULA

(a/b) × (c/d) = (a × c)/(b × d). Multiply numerators together and denominators together, then reduce to **lowest terms** (cancel common factors **before** multiplying to keep numbers small).

Product	Working	Result
3/5 × 10/21	(3 × 10)/(5 × 21) = 30/105, divide by 15	2/7
12/25 × 15/16	cancel 12 and 16 by 4, and 15 and 25 by 5: 3/4 × 3/5	9/20
2/3 × 45	(2 × 45)/3	30
1/2 × 1/4	1/8	1/8

- **'Of' means ×:** 2/3 of 45 litres = 2/3 × 45 = **30 litres**. 1/3 of a number is 12, so the number is 36 and 3/4 of it is **27**.
- **Mixed fractions:** convert to improper fractions first: 2 2/5 × 15 = 12/5 × 15 = **36**.
- Multiplying by a **proper fraction** gives a **smaller** number (3/4 × 2/3 = 1/2); multiplying by an **improper fraction** or mixed number gives a **larger** number.
- The area of a rectangle 3/4 m by 2/3 m is 3/4 × 2/3 = **1/2 m^2**.

Picture: shade 3/4 of a square, then shade 2/3 of that part: the doubly shaded part is 6/12 = 1/2 of the whole, which shows that multiplying fractions is a **fraction of a fraction**.

8.2 Reciprocals

DEFINITION

The **reciprocal** (multiplicative inverse) of a/b is **b/a**. A number times its reciprocal is **1**. The reciprocal of 7 is 1/7; of 2/3 is 3/2; of 1 3/4 = 7/4 is 4/7. **Zero has no reciprocal.**

Reciprocal of (2/3 + 1/2): first add, 4/6 + 3/6 = 7/6, so the reciprocal is **6/7**.

8.3 Dividing Fractions

FORMULA

Dividing by a fraction = multiplying by its reciprocal: a/b ÷ c/d = a/b × d/c.

Division	Meaning	Result
3 ÷ 1/8	how many 1/8 in 3?	3 × 8 = 24 pieces
7 ÷ 1 3/4	7 ÷ 7/4 = 7 × 4/7	4
2/3 ÷ 4	2/3 × 1/4	1/6
4 ÷ 2/3	4 × 3/2	6
3/4 ÷ 2/3	3/4 × 3/2	9/8 = 1 1/8

- Dividing by a number **less than 1** makes the result **bigger** (3/4 ÷ 2/3 = 9/8 > 3/4); dividing by a number greater than 1 makes it smaller.
- To divide a **whole number** by a fraction, ask ‘how many of the fraction fit?’: a 7 1/2 m cloth cut into pieces of 3/4 m gives 15/2 ÷ 3/4 = 15/2 × 4/3 = **10 pieces**.

8.4 Word Problems

Problem	Working	Answer
A car runs 15 km per litre. Distance on 2 2/5 L?	15 × 12/5	36 km
Ribbon of 3 m cut into pieces of 1/8 m	3 ÷ 1/8	24 pieces
Fraction of an hour in 18 minutes	18/60	3/10
Fraction of a day in 6 hours	6/24	1/4
3/5 of a 40 kg load	3/5 × 40	24 kg

REMEMBER

To **multiply**, no common denominator is needed. To **divide**, flip only the **second** fraction (the divisor), never the first.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Multiply 3/5 × 10/21 and give the answer in lowest terms.	2/7.
Find 2/3 of 45 litres.	30 litres.
How many pieces of 1/8 m can be cut from 3 m of ribbon?	24.
Define a reciprocal. Reciprocal of 2/3?	The number which, multiplied by it, gives 1; 3/2.
Divide 7 by 1 3/4.	7 × 4/7 = 4.

Likely question	Model answer
Which is larger, 3/4 × 2/3 or 3/4 ÷ 2/3?	The division, 9/8, as the divisor is less than 1.
Divide 2/3 by 4 and 4 by 2/3.	1/6 and 6.
A car goes 15 km per litre. How far on 2 2/5 litres?	36 km.
What fraction of an hour is 18 minutes?	3/10.
Does zero have a reciprocal?	No.

★ KEY TAKEAWAYS

- **Multiply:** numerator × numerator, denominator × denominator; ‘of’ means ×.
- **Reciprocal** of a/b is b/a; the product is 1; zero has none.
- **Divide by a fraction = multiply by its reciprocal.**
- Multiplying by a number less than 1 shrinks; **dividing by a number less than 1 enlarges**.
- Convert mixed fractions to **improper** before multiplying or dividing.